

❏ Concept of Windows Deployment Services (WDS)

❏ What is WDS?

Windows Deployment Services (WDS) is a server role in Windows Server that allows network-based installation of Windows operating systems to client computers without requiring physical media (DVDs or USBs). It is especially useful in enterprise or lab environments for mass deployment of Windows images.

❏ Key Components of WDS

Component	Description
PXE Boot	Preboot Execution Environment – allows computers to boot from the network.
WDS Server	Hosts and deploys the OS images. Must be a domain member.
Boot Image	A Windows PE (Preinstallation Environment) image used to boot client machines.
Install Image	The actual Windows OS image that will be installed on the client.
Image Groups	Logical containers for organizing install images.

❏ How WDS Works (Step-by-Step)

1.
Client boots via PXE and sends a request on the network.
2.
WDS server responds, offering the boot image.
3.
The client **downloads the boot image** (Windows PE).
4.
A technician or script chooses the **install image**.
5.
The OS installs automatically or semi-automatically.

❏ Requirements

- Active Directory Domain Services (AD DS)
 - DHCP Server (can be on same or separate server)
 - DNS
 - NTFS Partition on the WDS server
 - Windows image files (WIM format)
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☒ Supported Image Types

- .wim – Windows Image Format
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.vhd – for booting virtual hard disks

- Custom captured images (from syspreped reference machines)

☒ Deployment Modes

Mode	Description
Unattended Installation	Uses answer files to fully automate installation.
Multicast Deployment	Efficiently deploys images to many computers simultaneously.
Capture Image	Used to create a .wim image from a reference PC.
Discover Image	Bootable image for machines that can't PXE boot.

☒ Tools Used with WDS

- DISM – Deployment Image Servicing and Management
 - Sysprep – Prepares an OS for imaging
 - Windows SIM – For creating unattended answer files
 - WAIK/ADK – Toolkit to manage deployments
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☒ Use Cases

- Large organizations deploying OS to many machines
- Automated lab setups

- Quickly restoring systems to baseline state
 - Reducing manual work in OS deployment
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☒ **Security Considerations**

- Restrict who can deploy images via PXE
- Use AD-integrated WDS for better control
- Protect unattended answer files with credentials